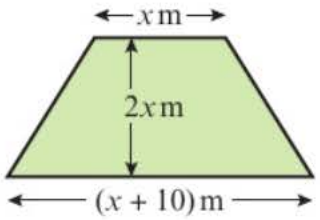
(P) **4** This trapezium has an area of $50\,m^2$.
Show that the height of the trapezium is equal to $5(\sqrt{5}-1)\,m$.



Problem-solving

Height must be positive. You will have to discard the negative solution of your quadratic equation.

Challenge

Given that x is positive, solve the equation

$$\frac{1}{x}+\frac{1}{x+2}=\frac{28}{195}$$

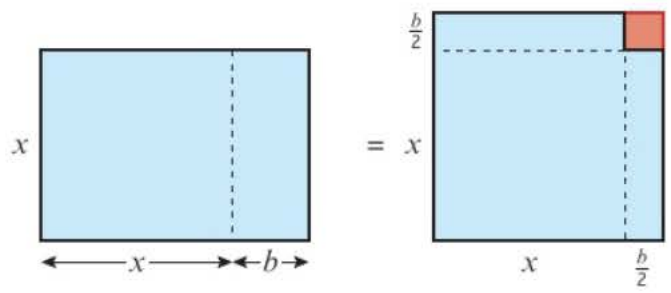
Hint Write the equation in the form $ax^2 + bx + c = 0$ before using the quadratic formula or factorising.

2.2 Completing the square

It is frequently useful to rewrite quadratic expressions by **completing the square:**

- $\boldsymbol{x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2}$

You can draw a diagram of this process when x and b are positive:



The original rectangle has been rearranged into the shape of a square with a smaller square missing. The two areas shaded blue are the same.

$$x^2 + bx = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$$

Example 4

Complete the square for the expressions:

a $x^2 + 8x$ **b** $x^2 - 3x$ **c** $2x^2 + 12x$

a $x^2 + 8x = (x + 4)^2 - 4^2$
$= (x + 4)^2 - 16$

Begin by halving the coefficient of x. Using the rule given above, $b = 8$ so $\frac{b}{2} = 4$.

b $x^2 - 3x = (x - \frac{3}{2})^2 - (\frac{3}{2})^2$
$= (x - \frac{3}{2})^2 - \frac{9}{4}$

Be careful if $\frac{b}{2}$ is a fraction. Here $\left(\frac{3}{2}\right)^2 = \frac{3^2}{2^2} = \frac{9}{4}$.

c $2x^2 - 12x = 2(x^2 - 6x)$
$= 2((x - 3)^2 - 3^2)$
$= 2((x - 3)^2 - 9)$
$= 2(x - 3)^2 - 18$

Here the coefficient of x^2 is 2, so take out a factor of 2. The other factor is in the form $(x^2 + bx)$ so you can use the rule to complete the square.

Expand the outer bracket by multiplying 2 by 9 to get your answer in this form.

Notation A quadratic expression in the form $p(x + q)^2 + r$ where p, q and r are real constants is in **completed square form**.

- $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a^2}\right)$

Example 5

Write $3x^2 + 6x + 1$ in the form $p(x + q)^2 + r$, where p, q and r are integers to be found.

$3x^2 + 6x + 1$
$= 3(x^2 + 2x) + 1$
$= 3((x + 1)^2 - 1^2) + 1$
$= 3(x + 1)^2 - 3 + 1$
$= 3(x + 1)^2 - 2$
So $p = 3$, $q = 1$ and $r = -2$.

Watch out This is an **expression**, so you can't divide every term by 3 without changing its value. Instead, you need to take a factor of 3 out of $3x^2 + 6x$.

You could also use the rule given above to complete the square for this expression, but it is safer to learn the method shown here.

Exercise 2C

1 Complete the square for the expressions:

a $x^2 + 4x$ **b** $x^2 - 6x$ **c** $x^2 - 16x$ **d** $x^2 + x$ **e** $x^2 - 14$

2 Complete the square for the expressions:

a $2x^2 + 16x$ **b** $3x^2 - 24x$ **c** $5x^2 + 20x$ **d** $2x^2 - 5x$ **e** $8x - 2x^2$

Hint In question **3d**, write the expression as $-4x^2 - 16x + 10$ then take a factor of -4 out of the first two terms to get $-4(x^2 + 4x) + 10$.

3 Write each of these expressions in the form $p(x + q)^2 + r$, where p, q and r are constants to be found:

a $2x^2 + 8x + 1$ **b** $5x^2 - 15x + 3$ **c** $3x^2 + 2x - 1$ **d** $10 - 16x - 4x^2$ **e** $2x - 8x^2 + 10$

E **4** Given that $x^2 + 3x + 6 = (x + a)^2 + b$, find the values of the constants a and b. **(2 marks)**

E **5** Write $2 + 0.8x - 0.04x^2$ in the form $A - B(x + C)^2$, where A, B and C are constants to be determined. **(3 marks)**

Example 6

Solve the equation $x^2 + 8x + 10 = 0$ by completing the square.
Give your answers in surd form.

$x^2 + 8x + 10 = 0$ — Check coefficient of $x^2 = 1$.
$x^2 + 8x = -10$ — Subtract 10 to get the LHS in the form $x^2 + bx$.
$(x + 4)^2 - 4^2 = -10$ — Complete the square for $x^2 + 8x$.
$(x + 4)^2 = -10 + 16$ — Add 4^2 to both sides.
$(x + 4)^2 = 6$
$(x + 4) = \pm\sqrt{6}$ — Take square roots of both sides.
$x = -4 \pm \sqrt{6}$ — Subtract 4 from both sides.
So the solutions are
$x = -4 + \sqrt{6}$ and $x = -4 - \sqrt{6}$. — Leave your answer in surd form.

Example 7

Solve the equation $2x^2 - 8x + 7 = 0$. Give your answers in surd form.

$2x^2 - 8x + 7 = 0$

$x^2 - 4x + \frac{7}{2} = 0$

$x^2 - 4x = -\frac{7}{2}$

$(x - 2)^2 - 2^2 = -\frac{7}{2}$

$(x - 2)^2 = -\frac{7}{2} + 4$

$(x - 2)^2 = \frac{1}{2}$

$x - 2 = \pm\sqrt{\frac{1}{2}}$

$x = 2 \pm \frac{1}{\sqrt{2}}$

So the roots are

$x = 2 + \frac{1}{\sqrt{2}}$ and $x = 2 - \frac{1}{\sqrt{2}}$

Problem-solving

This is an **equation** so you can divide every term by the same constant. Divide by 2 to get x^2 on its own. The right-hand side is 0 so it is unchanged.

Complete the square for $x^2 - 4x$.

Add 2^2 to both sides.

Take square roots of both sides.

Add 2 to both sides.

Online Use your calculator to check solutions to quadratic equations quickly.

Exercise 2D

1 Solve these quadratic equations by completing the square. Leave your answers in surd form.

a $x^2 + 6x + 1 = 0$ **b** $x^2 + 12x + 3 = 0$ **c** $x^2 + 4x - 2 = 0$ **d** $x^2 - 10x = 5$

2 Solve these quadratic equations by completing the square. Leave your answers in surd form.

a $2x^2 + 6x - 3 = 0$ **b** $5x^2 + 8x - 2 = 0$ **c** $4x^2 - x - 8 = 0$ **d** $15 - 6x - 2x^2 = 0$

E **3** $x^2 - 14x + 1 = (x + p)^2 + q$, where p and q are constants.

a Find the values of p and q. **(2 marks)**

b Using your answer to part **a**, or otherwise, show that the solutions to the equation $x^2 - 14x + 1 = 0$ can be written in the form $r \pm s\sqrt{3}$, where r and s are constants to be found. **(2 marks)**

E/P **4** By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. **(4 marks)**

Problem-solving

Follow the same steps as you would if the coefficients were numbers.

Challenge

a Show that the solutions to the equation $ax^2 + 2bx + c = 0$ are given by $x = -\frac{b}{a} \pm \sqrt{\frac{b^2 - ac}{a^2}}$.

b Hence, or otherwise, show that the solutions to the equation $ax^2 + bx + c = 0$ can be written as $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Hint Start by dividing the whole equation by a.

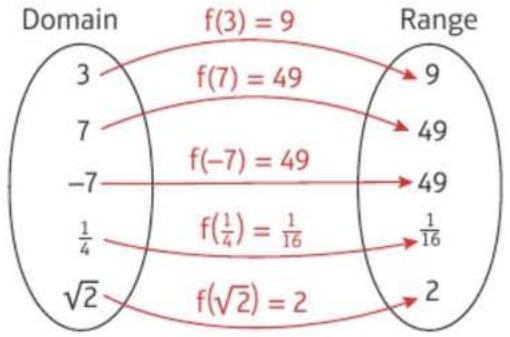
Links You can use this method to prove the quadratic formula. → Section 7.4

2.3 Functions

A function is a mathematical relationship that maps each value of a set of inputs to a single output. The notation f(x) is used to represent a function of x.

- **The set of possible inputs for a function is called the domain.**
- **The set of possible outputs of a function is called the range.**

This diagram shows how the function $f(x) = x^2$ maps five values in its domain to values in its range.



- **The roots of a function are the values of x for which f(x) = 0.**

Example 8

The functions f and g are given by $f(x) = 2x - 10$ and $g(x) = x^2 - 9$, $x \in \mathbb{R}$.

a Find the values of f(5) and g(10).

b Find the value of x for which $f(x) = g(x)$.

Notation If the input of a function, x, can be any real number the domain can be written as $x \in \mathbb{R}$.

The symbol $\in$ means 'is a member of' and the symbol $\mathbb{R}$ represents the real numbers.

a $f(5) = 2(5) - 10 = 10 - 10 = 0$
$g(10) = (10)^2 - 9 = 100 - 9 = 91$

To find f(5), substitute $x = 5$ into the function f(x).

b
$f(x) = g(x)$
$2x - 10 = x^2 - 9$
$x^2 - 2x + 1 = 0$
$(x - 1)^2 = 0$
$x = 1$

Set f(x) equal to g(x) and solve for x.

Example 9

The function f is defined as $f(x) = x^2 + 6x - 5$, $x \in \mathbb{R}$.

a Write f(x) in the form $(x + p)^2 + q$.

b Hence, or otherwise, find the roots of f(x), leaving your answers in surd form.

c Write down the minimum value of f(x), and state the value of x for which it occurs.

a
$f(x) = x^2 + 6x - 5$
$= (x + 3)^2 - 9 - 5$
$= (x + 3)^2 - 14$

Complete the square for $x^2 + 6x$ and then simplify the expression.

b
$f(x) = 0$
$(x + 3)^2 - 14 = 0$
$(x + 3)^2 = 14$
$x + 3 = \pm\sqrt{14}$
$x = -3 \pm \sqrt{14}$

f(x) has two roots:
$-3 + \sqrt{14}$ and $-3 - \sqrt{14}$.

To find the root(s) of a function, set it equal to zero.

You can solve this equation directly. Remember to write ± when you take square roots of both sides.

4 The functions p and q are given by $p(x) = x^2 - 3x$ and $q(x) = 2x - 6$, $x \in \mathbb{R}$.
Find the two values of x for which $p(x) = q(x)$.

5 The functions f and g are given by $f(x) = 2x^3 + 30x$ and $g(x) = 17x^2$, $x \in \mathbb{R}$.
Find the three values of x for which $f(x) = g(x)$.

E **6** The function f is defined as $f(x) = x^2 - 2x + 2$, $x \in \mathbb{R}$.

a Write $f(x)$ in the form $(x + p)^2 + q$, where p and q are constants to be found. **(2 marks)**

b Hence, or otherwise, explain why $f(x) > 0$ for all values of x, and find the minimum value of $f(x)$. **(1 mark)**

7 Find all roots of the following functions:

a $f(x) = x^6 + 9x^3 + 8$

b $g(x) = x^4 - 12x^2 + 32$

c $h(x) = 27x^6 + 26x^3 - 1$

d $j(x) = 32x^{10} - 33x^5 + 1$

e $k(x) = x - 7\sqrt{x} + 10$

f $m(x) = 2x^{\frac{2}{3}} + 2x^{\frac{1}{3}} - 12$

Hint The function in part **b** has four roots.

E/P **8** The function f is defined as $f(x) = 3^{2x} - 28(3^x) + 27$, $x \in \mathbb{R}$.

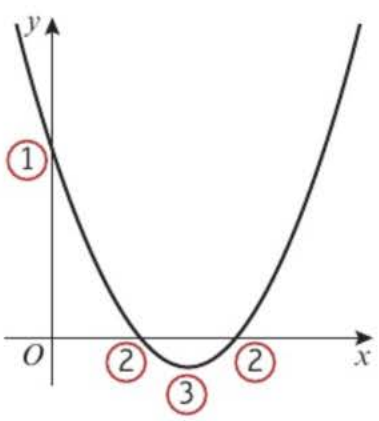
a Write $f(x)$ in the form $(3^x - a)(3^x - b)$, where a and b are real constants. **(2 marks)**

b Hence find the two roots of $f(x)$. **(2 marks)**

Problem-solving
Consider $f(x)$ as a function of a function.

2.4 Quadratic graphs

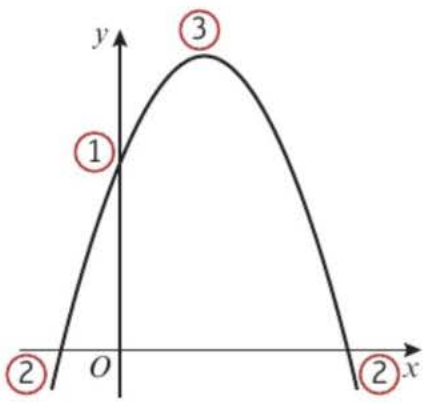
When $f(x) = ax^2 + bx + c$, the graph of $y = f(x)$ has a curved shape called a parabola.
You can sketch a quadratic graph by identifying key features.
The coefficient of x^2 determines the overall shape of the graph.
When a is positive the parabola will have this shape: ∪
When a is negative the parabola will have this shape: ∩



(1) The graph crosses the y-axis when $x = 0$. The y-coordinate is equal to c.

(2) The graph crosses the x-axis when $y = 0$. The x-coordinates are roots of the function $f(x)$.

(3) Quadratic graphs have one turning point. This can be a minimum or a maximum. Since a parabola is symmetrical, the turning point and line of symmetry are half-way between the two roots.



■ **You can find the coordinates of the turning point of a quadratic graph by completing the square. If $f(x) = a(x + p)^2 + q$, the graph of $y = f(x)$ has a turning point at $(-p, q)$.**

Links The graph of $y = a(x + p)^2 + q$ is a translation of the graph of $y = ax^2$ by $\begin{pmatrix} -p \\ q \end{pmatrix}$. → Section 4.5

Example 11

Sketch the graph of $y = x^2 - 5x + 4$, and find the coordinates of its turning point.

As $a = 1$ is positive, the graph has a $\vee$ shape and a minimum point.

When $x = 0$, $y = 4$, so the graph crosses the y-axis at $(0, 4)$.

When $y = 0$,

$$x^2 - 5x + 4 = 0$$

$$(x - 1)(x - 4) = 0$$

$x = 1$ or $x = 4$, so the graph crosses the x-axis at $(1, 0)$ and $(4, 0)$.

Completing the square:

$$x^2 - 5x + 4 = \left(x - \tfrac{5}{2}\right)^2 - \tfrac{25}{4} + 4$$

$$= \left(x - \tfrac{5}{2}\right)^2 - \tfrac{9}{4}$$

So the minimum point has coordinates $\left(\tfrac{5}{2}, -\tfrac{9}{4}\right)$.

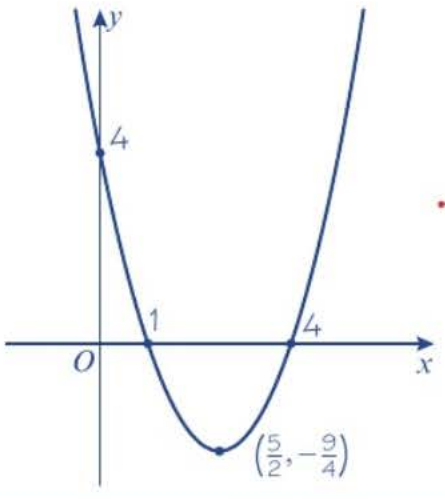
Alternatively, the minimum occurs when x is half-way between 1 and 4,

so $x = \dfrac{1 + 4}{2} = \tfrac{5}{2}$

$$y = \left(\tfrac{5}{2}\right)^2 - 5 \times \left(\tfrac{5}{2}\right) + 4 = -\tfrac{9}{4}$$

so the minimum has coordinates $\left(\tfrac{5}{2}, -\tfrac{9}{4}\right)$.

The sketch of the graph is:



Use the coefficient of x^2 to determine the general shape of the graph.

This example factorises, but you may need to use the quadratic formula or complete the square.

Complete the square to find the coordinates of the turning point.

Watch out If you use symmetry to find the x-coordinate of the minimum point, you need to substitute this value into the equation to find the y-coordinate of the minimum point.

You could use a graphic calculator or substitute some values to check your sketch.

When $x = 5$, $y = 5^2 - 5 \times 5 + 4 = 4$.

Online Explore how the graph of $y = (x + p)^2 + q$ changes as the values of p and q change using GeoGebra.

Example 12

Sketch the graph of $y = 4x - 2x^2 - 3$. Find the coordinates of its turning point and write down the equation of its line of symmetry.

As $a = -2$ is negative, the graph has a $\cap$ shape and a maximum point.

When $x = 0$, $y = -3$, so the graph crosses the y-axis at $(0, -3)$.

When $y = 0$,

$-2x^2 + 4x - 3 = 0$

Using the quadratic formula,

$$x = \frac{-4 \pm \sqrt{4^2 - 4(-2)(-3)}}{2 \times (-2)}$$

$$x = \frac{-4 \pm \sqrt{-8}}{-4}$$

There are no real solutions, so the graph does not cross the x-axis.

Completing the square:

$-2x^2 + 4x - 3$

$= -2(x^2 - 2x) - 3$

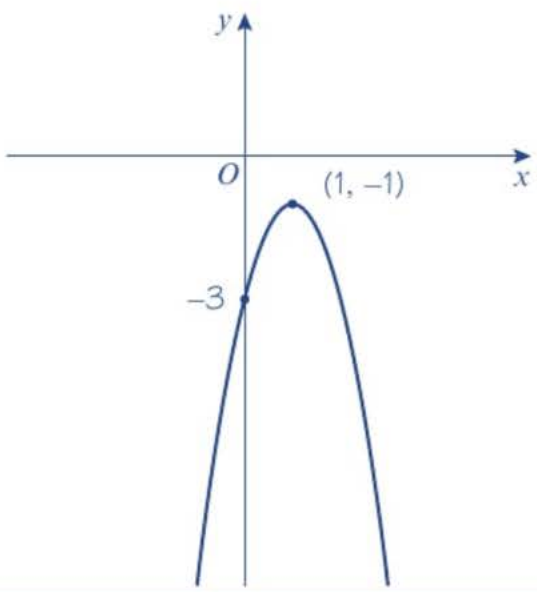
$= -2((x - 1)^2 - 1) - 3$

$= -2(x - 1)^2 + 2 - 3$

$= -2(x - 1)^2 - 1$

So the maximum point has coordinates $(1, -1)$.

The line of symmetry is vertical and goes through the maximum point. It has the equation $x = 1$.



It's easier to see that $a < 0$ if you write the equation in the form $y = -2x^2 + 4x - 3$.

$a = -2$, $b = 4$ and $c = -3$

You would need to square root a negative number to evaluate this expression. Therefore this equation has no real solutions.

The coefficient of x^2 is -2 so take out a factor of -2

Watch out A sketch graph does not need to be plotted exactly or drawn to scale. However you should:

- draw a smooth curve by hand
- identify any relevant key points (such as intercepts and turning points)
- label your axes.

Exercise 2F

1 Sketch the graphs of the following equations. For each graph, show the coordinates of the point(s) where the graph crosses the coordinate axes, and write down the coordinate of the turning point and the equation of the line of symmetry.

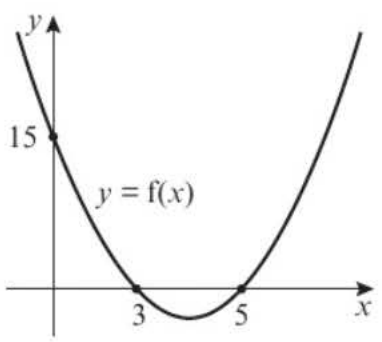
a $y = x^2 - 6x + 8$ **b** $y = x^2 + 2x - 15$ **c** $y = 25 - x^2$ **d** $y = x^2 + 3x + 2$

e $y = -x^2 + 6x + 7$ **f** $y = 2x^2 + 4x + 10$ **g** $y = 2x^2 + 7x - 15$ **h** $y = 6x^2 - 19x + 10$

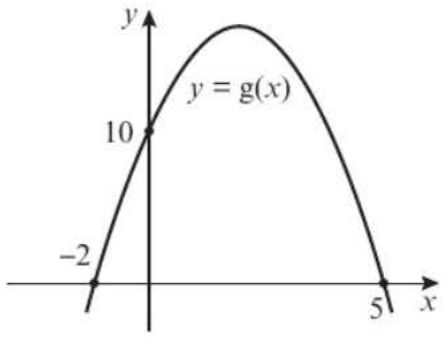
i $y = 4 - 7x - 2x^2$ **j** $y = 0.5x^2 + 0.2x + 0.02$

(P) **2** These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$.
Find the values of a, b and c for each function.

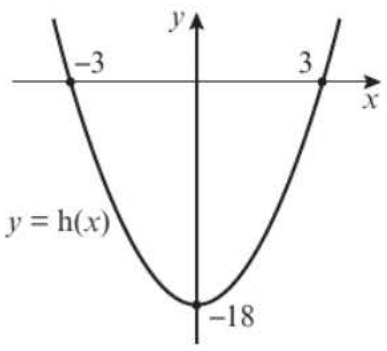
a



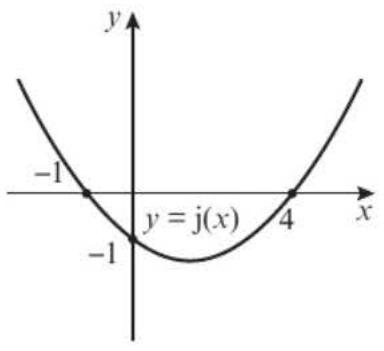
b



c



d



Problem-solving

Check your answers by substituting values into the function. In part **c** the graph passes through (0, –18), so h(0) should be –18.

(E/P) **3** The graph of $y = ax^2 + bx + c$ has a minimum at (5, –3) and passes through (4, 0).
Find the values of a, b and c. **(3 marks)**

2.5 The discriminant

If you square any real number, the result is greater than or equal to 0. This means that if y is negative, $\sqrt{y}$ cannot be a real number. Look at the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

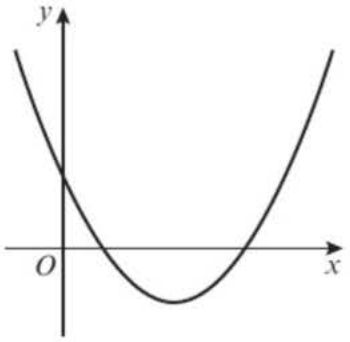
If the value under the square root sign is negative, x cannot be a real number and there are no real solutions. If the value under the square root is equal to 0, both solutions will be the same.

- **For the quadratic function $f(x) = ax^2 + bx + c$, the expression $b^2 - 4ac$ is called the discriminant. The value of the discriminant shows how many roots $f(x)$ has:**
  - **If $b^2 - 4ac > 0$ then $f(x)$ has two distinct real roots.**
  - **If $b^2 - 4ac = 0$ then $f(x)$ has one repeated root.**
  - **If $b^2 - 4ac < 0$ then $f(x)$ has no real roots.**

You can use the discriminant to check the shape of sketch graphs.

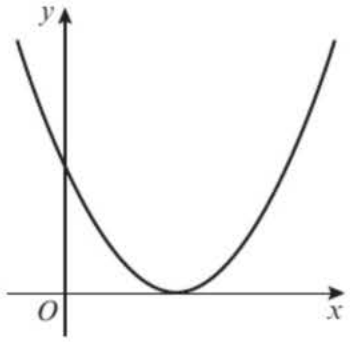
Below are some graphs of $y = f(x)$ where $f(x) = ax^2 + bx + c$.

$a > 0$



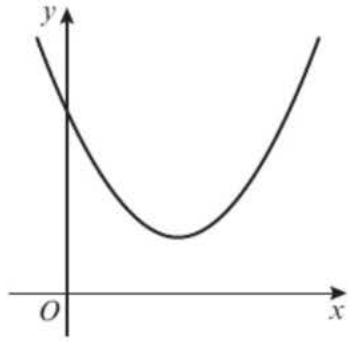
$b^2 - 4ac > 0$

Two distinct real roots



$b^2 - 4ac = 0$

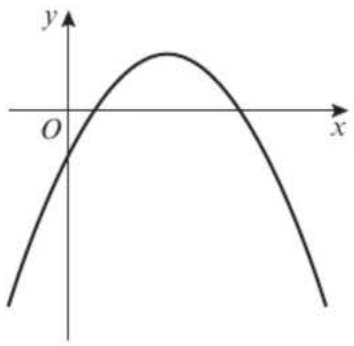
One repeated root

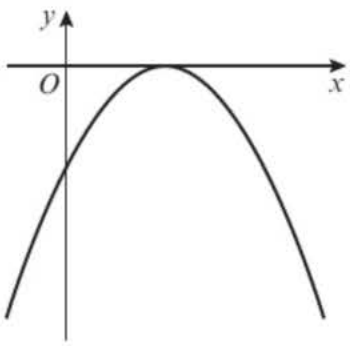


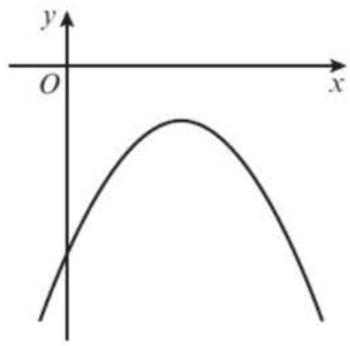
$b^2 - 4ac < 0$

No real roots

$a < 0$







Example 13

Find the values of k for which $f(x) = x^2 + kx + 9$ has equal roots.

$x^2 + kx + 9 = 0$
Here $a = 1$, $b = k$ and $c = 9$
For equal roots, $b^2 - 4ac = 0$
$k^2 - 4 \times 1 \times 9 = 0$
$k^2 - 36 = 0$
$k^2 = 36$
so $k = \pm 6$

Problem-solving

Use the condition given in the question to write a statement about the discriminant.

Substitute for a, b and c to get an equation with one unknown.

Solve to find the values of k.

Example 14

Find the range of values of k for which $x^2 + 4x + k = 0$ has two distinct real solutions.

$x^2 + 4x + k = 0$
Here $a = 1$, $b = 4$ and $c = k$.
For two real solutions, $b^2 - 4ac > 0$
$4^2 - 4 \times 1 \times k > 0$
$16 - 4k > 0$
$16 > 4k$
$4 > k$
So $k < 4$

This statement involves an inequality, so your answer will also be an inequality.

For any value of k less than 4, the equation will have 2 distinct real solutions.

Online Explore how the value of the discriminant changes with k using GeoGebra.

Exercise 2G

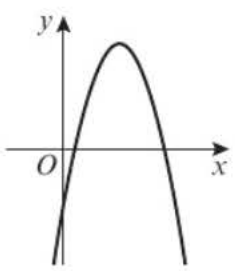
1 a Calculate the value of the discriminant for each of these five functions:

i $f(x) = x^2 + 8x + 3$ **ii** $g(x) = 2x^2 - 3x + 4$ **iii** $h(x) = -x^2 + 7x - 3$

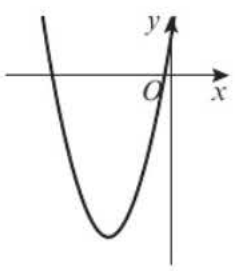
iv $j(x) = x^2 - 8x + 16$ **v** $k(x) = 2x - 3x^2 - 4$

b Using your answers to part **a**, match the same five functions to these sketch graphs.

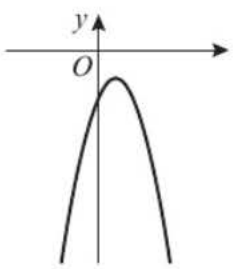
i



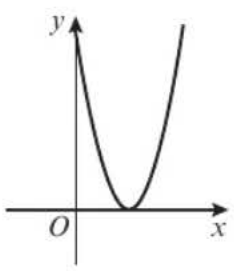
ii



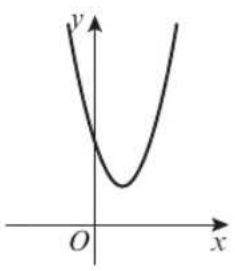
iii



iv



v



E/P **2** Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions. **(2 marks)**

E/P **3** Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution. **(2 marks)**

E/P **4** Given that the function $f(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s. **(2 marks)**

E/P **5** Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions. **(2 marks)**

E/P **6** The function $g(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.

a Find the value of p. **(2 marks)**

b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$. **(2 marks)**

E/P **7** $h(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.

a Find the discriminant of $h(x)$ in terms of k. **(3 marks)**

b Hence or otherwise, prove that $h(x)$ has two distinct real roots for all values of k. **(3 marks)**

Problem-solving

If a question part says 'hence or otherwise' it is usually easier to use your answer to the previous question part.

Challenge

a Prove that, if the values of a and c are given and non-zero, it is always possible to choose a value of b so that $f(x) = ax^2 + bx + c$ has distinct real roots.

b Is it always possible to choose a value of b so that $f(x)$ has equal roots? Explain your answer.

2.6 Modelling with quadratics

A **mathematical model** is a mathematical description of a real-life situation. Mathematical models use the language and tools of mathematics to represent and explore real-life patterns and relationships, and to predict what is going to happen next.

Models can be simple or complicated, and their results can be approximate or exact. Sometimes a model is only valid under certain circumstances, or for a limited range of inputs. You will learn more about how models involve simplifications and assumptions in Statistics and Mechanics.

Quadratic functions can be used to model and explore a range of practical contexts, including projectile motion.